

Work, Energy & Power: Session 5

Advanced Olympiad Challenge Sheet

Focus: Conservative vs. Non-Conservative Fields, Curls, Central Potentials, and Frame Transformations

Session 5 Theoretical Anchors

In Session 5, we transition from scalar work definitions to the vector field representation of forces. Key mathematical theorems from this lecture include:

- **The Path Independence Theorem:** A force field $\vec{F}$ is conservative if and only if $\oint \vec{F} \cdot d\vec{s} = 0$ along any closed loop, which guarantees that $W_{A \rightarrow B} = -\Delta U$ is path-independent.
- **The Local Differential Test (Curl):** In a 3D coordinate space, $\vec{F}$ is conservative if and only if $\vec{\nabla} \times \vec{F} = \vec{0}$. This corresponds to three simultaneous partial differential constraints:

$$\frac{\partial F_z}{\partial y} = \frac{\partial F_y}{\partial z}, \quad \frac{\partial F_x}{\partial z} = \frac{\partial F_z}{\partial x}, \quad \frac{\partial F_y}{\partial x} = \frac{\partial F_x}{\partial y}$$

- **Central Force Invariance:** Any central force $\vec{F} = f(r)\hat{r}$ has zero curl and is therefore conservative. Its potential energy is expressed as $U(r) = -\int f(r)dr$.

Challenge 1: Path Dependence of a 2D Non-Conservative Vector Field

The Setup: A force field in the XY -plane is given by:

$$\vec{F}(x, y) = \beta y^2 \hat{i} + \alpha x^2 \hat{j}$$

where α and β are non-zero constant parameters.

1. Calculate the work done by this force in moving a particle from the origin $O(0,0)$ to the point $C(1,1)$ along the straight line path $y = x$.
2. Calculate the work done in moving the same particle between the same two points along the parabolic path $y = x^2$.
3. Prove using the 2D local differential test (curl) that this force field is non-conservative for all non-zero values of α and β .

Challenge 1: Progressive Guidance

- **Hint 1 (Line Path):** Along $y = x$, we have $dy = dx$. Substitute both $y = x$ and $dy = dx$ into the work line integral $W = \int (F_x dx + F_y dy)$ to make it a single integral with respect to x from 0 to 1.
- **Hint 2 (Parabola Path):** Along $y = x^2$, differentiate to find $dy = 2x dx$. Substitute $y = x^2$ and $dy = 2x dx$ into the integrand. Keep your integration limits with respect to x as $[0, 1]$.
- **Hint 3 (Differential Test):** For a 2D force field $\vec{F} = F_x \hat{i} + F_y \hat{j}$, the only curl component is in the $\hat{k}$ direction. Evaluate $\frac{\partial F_y}{\partial x}$ and $\frac{\partial F_x}{\partial y}$. For the force to be conservative, these two partial derivatives must be identically equal everywhere in space.

Challenge 1: Analytical Derivation**1. Work along the straight line $y = x$ ($dy = dx$):**

$$W_{\text{line}} = \int_{(0,0)}^{(1,1)} (\beta y^2 dx + \alpha x^2 dy) = \int_0^1 (\beta x^2 dx + \alpha x^2 dx) = (\beta + \alpha) \int_0^1 x^2 dx = \frac{\alpha + \beta}{3}$$

2. Work along the parabola $y = x^2$ ($dy = 2x dx$):

$$W_{\text{parabola}} = \int_0^1 [\beta (x^2)^2 dx + \alpha x^2 (2x dx)] = \int_0^1 [\beta x^4 + 2\alpha x^3] dx = \left[\beta \frac{x^5}{5} + 2\alpha \frac{x^4}{4} \right]_0^1 = \frac{\alpha}{2} + \frac{\beta}{5}$$

3. Differential Test: Evaluate the partial derivatives of the components:

$$F_x = \beta y^2 \implies \frac{\partial F_x}{\partial y} = 2\beta y, \quad F_y = \alpha x^2 \implies \frac{\partial F_y}{\partial x} = 2\alpha x$$

For the curl to vanish, we must have:

$$2\alpha x = 2\beta y \quad \text{for all } (x, y)$$

This condition only holds at the origin or if $\alpha = \beta = 0$. Since α and β are non-zero constants, the partial derivatives are not identical everywhere. Thus, the curl is non-zero, proving the field is ****strictly non-conservative****.

Challenge 2: The 3D Vector Curl and Parameterized Potential Energy

The Setup: A three-dimensional force field acting on a particle is given by the vector function:

$$\vec{F}(x, y, z) = (y^2z^3 + \alpha x^2y)\hat{i} + (2xyz^3 + x^3)\hat{j} + (3xy^2z^2 + \gamma z)\hat{k}$$

where α and γ are constants.

1. Find the unique value of the parameter α that makes this force field conservative.
2. Using this value of α , find the potential energy function $U(x, y, z)$ of the system, taking the origin $O(0, 0, 0)$ as the reference point where $U(0, 0, 0) = 0$.

Challenge 2: Progressive Guidance

- **Hint 1 (Symmetry of Partial Derivatives):** Write down the components: $F_x = y^2z^3 + \alpha x^2y$, $F_y = 2xyz^3 + x^3$, and $F_z = 3xy^2z^2 + \gamma z$. Evaluate $\frac{\partial F_y}{\partial x}$ and $\frac{\partial F_x}{\partial y}$ and set them equal to solve for α .
- **Hint 2 (Double Checking Other Pairs):** Ensure that your value of α also satisfies $\frac{\partial F_z}{\partial y} = \frac{\partial F_y}{\partial z}$ and $\frac{\partial F_x}{\partial z} = \frac{\partial F_z}{\partial x}$.
- **Hint 3 (Potential Energy Integration):** Recall that $dU = -\vec{F} \cdot d\vec{s} = -(F_x dx + F_y dy + F_z dz)$. Look for perfect differentials or perform path-independent integration. For instance, integrate along the path $(0, 0, 0) \rightarrow (x, 0, 0) \rightarrow (x, y, 0) \rightarrow (x, y, z)$.

Challenge 2: Analytical Derivation

1. Finding α via Curl Test: Let's compute the cross partial derivatives:

$$\frac{\partial F_x}{\partial y} = 2yz^3 + \alpha x^2 \quad \text{and} \quad \frac{\partial F_y}{\partial x} = 2yz^3 + 3x^2$$

For the z-component of the curl to vanish, these must be equal:

$$2yz^3 + \alpha x^2 = 2yz^3 + 3x^2 \implies \alpha = 3$$

Let's verify the remaining pairs to ensure complete consistency:

$$\frac{\partial F_z}{\partial y} = 6xyz^2, \quad \frac{\partial F_y}{\partial z} = 6xyz^2 \quad (\text{Equal!})$$

$$\frac{\partial F_x}{\partial z} = 3y^2z^2, \quad \frac{\partial F_z}{\partial x} = 3y^2z^2 \quad (\text{Equal!})$$

Thus, the force is conservative if and only if $\alpha = 3$ (the value of γ does not affect conservativeness since γz is a 1D force).

2. Deriving the Potential Energy Function $U(x, y, z)$: We write $dU = -(F_x dx + F_y dy + F_z dz)$:

$$dU = -[(y^2z^3 + 3x^2y)dx + (2xyz^3 + x^3)dy + (3xy^2z^2 + \gamma z)dz]$$

Grouping the terms to identify perfect differentials:

$$dU = -[(y^2z^3 dx + 2xyz^3 dy + 3xy^2z^2 dz) + (x^3 dy + \gamma z dz)]$$

Notice that:

$$d(xy^2z^3) = y^2z^3 dx + 2xyz^3 dy + 3xy^2z^2 dz$$

$$d(x^3y) = 3x^2y dx + x^3 dy$$

$$d\left(\frac{\gamma z^2}{2}\right) = \gamma z dz$$

Thus, the entire differential simplifies to:

$$dU = -d\left(xy^2z^3 + x^3y + \frac{\gamma z^2}{2}\right)$$

Integrating from the origin reference $U(0, 0, 0) = 0$ to $U(x, y, z)$:

$$U(\mathbf{x}, \mathbf{y}, \mathbf{z}) = -\left(\mathbf{x}\mathbf{y}^2\mathbf{z}^3 + \mathbf{x}^3\mathbf{y} + \frac{\gamma\mathbf{z}^2}{2}\right)$$

Challenge 3: Central Forces and Spherically Symmetric Potentials

The Setup: A central force field in 3D space is defined by:

$$\vec{F}(\vec{r}) = f(r)\hat{r}$$

where $r = \sqrt{x^2 + y^2 + z^2}$ is the radial distance from the origin, and $\hat{r} = \frac{\vec{r}}{r} = \frac{x\hat{i} + y\hat{j} + z\hat{k}}{r}$ is the radial unit vector.

1. Rigorously prove that any such central force field has zero curl ($\vec{\nabla} \times \vec{F} = \vec{0}$) in Cartesian coordinates, establishing that all central forces are conservative.
2. For the specific gravitational/electrostatic inverse-square attraction where $f(r) = -\frac{k}{r^2}$ ($k > 0$), calculate the potential energy function $U(r)$ choosing the reference point at infinity ($U(\infty) = 0$).

Challenge 3: Progressive Guidance

- **Hint 1 (Chain Rule for r):** Note that $\frac{\partial r}{\partial x} = \frac{x}{r}$, $\frac{\partial r}{\partial y} = \frac{y}{r}$, and $\frac{\partial r}{\partial z} = \frac{z}{r}$. Express the Cartesian components of the force as $F_x = f(r)\frac{x}{r}$, $F_y = f(r)\frac{y}{r}$, and $F_z = f(r)\frac{z}{r}$.
- **Hint 2 (Evaluating the Curl Components):** Evaluate the partial derivative $\frac{\partial F_y}{\partial x}$ using the product and chain rules. Show that it is symmetric with respect to x and y , which means $\frac{\partial F_y}{\partial x} = \frac{\partial F_x}{\partial y}$.
- **Hint 3 (Potential Integration):** Since the field is spherically symmetric, the line integral simplifies to a radial integration: $U(r) = -\int_{\infty}^r f(r')dr'$.

Challenge 3: Analytical Derivation

1. **Proving $\vec{\nabla} \times \vec{F} = \vec{0}$:** We express $F_x = x\frac{f(r)}{r}$ and $F_y = y\frac{f(r)}{r}$. Let's find the cross partials:

$$\frac{\partial F_y}{\partial x} = \frac{\partial}{\partial x} \left[y \frac{f(r)}{r} \right] = y \frac{\partial}{\partial x} \left[\frac{f(r)}{r} \right]$$

By the chain rule, since $r = \sqrt{x^2 + y^2 + z^2} \implies \frac{\partial r}{\partial x} = \frac{x}{r}$:

$$\frac{\partial}{\partial x} \left[\frac{f(r)}{r} \right] = \frac{d}{dr} \left[\frac{f(r)}{r} \right] \frac{\partial r}{\partial x} = \left[\frac{f'(r)r - f(r)}{r^2} \right] \frac{x}{r} = x \left[\frac{rf'(r) - f(r)}{r^3} \right]$$

Thus:

$$\frac{\partial F_y}{\partial x} = xy \left[\frac{rf'(r) - f(r)}{r^3} \right]$$

By exact symmetry, evaluating $\frac{\partial F_x}{\partial y}$:

$$\frac{\partial F_x}{\partial y} = \frac{\partial}{\partial y} \left[x \frac{f(r)}{r} \right] = x \frac{d}{dr} \left[\frac{f(r)}{r} \right] \frac{\partial r}{\partial y} = xy \left[\frac{rf'(r) - f(r)}{r^3} \right]$$

Since $\frac{\partial F_y}{\partial x} = \frac{\partial F_x}{\partial y}$, the z -component of the curl is zero. By cyclic permutation, the x - and y -components of the curl also vanish identically. Thus, **any central force field is conservative**.

2. **Finding the Potential Energy for $f(r) = -\frac{k}{r^2}$:** Using the definition of potential energy with reference at infinity:

$$U(r) = -\int_{\infty}^r f(r')dr' = -\int_{\infty}^r \left(-\frac{k}{r'^2} \right) dr' = -k \left[\frac{1}{r'} \right]_{\infty}^r = -\frac{k}{r}$$

Challenge 4: Non-Conservative Circulating Force on a Logarithmic Spiral

The Setup: A circulating, non-conservative force acts in the horizontal XY -plane:

$$\vec{F} = -k(y\hat{i} - x\hat{j})$$

where k is a positive constant. A particle is constrained to move along a logarithmic spiral path given in polar coordinates by:

$$r(\theta) = R_0 e^{-a\theta}$$

where R_0 and a are positive constants, as the angle θ sweeps from $\theta = 0$ to $\theta = 2\pi$.

1. Convert the force vector $\vec{F}$ into polar coordinates (using $\hat{r}$ and $\hat{\theta}$).
2. Calculate the total work done by this force during the spiral trajectory.

Challenge 4: Progressive Guidance

- **Hint 1 (Polar Conversions):** Recall the transformations: $x = r \cos \theta$, $y = r \sin \theta$, and the unit vectors:

$$\hat{r} = \cos \theta \hat{i} + \sin \theta \hat{j}, \quad \hat{\theta} = -\sin \theta \hat{i} + \cos \theta \hat{j}$$

Substitute x and y into $\vec{F}$ and group them to match the $\hat{\theta}$ unit vector.

- **Hint 2 (Polar Displacement):** The infinitesimal displacement vector in polar coordinates is $d\vec{s} = dr\hat{r} + r d\theta \hat{\theta}$. Find the dot product $dW = \vec{F} \cdot d\vec{s}$.
- **Hint 3 (Spiral Integration):** Substitute the path equation $r = R_0 e^{-a\theta}$ into your expression for dW . Integrate the resulting exponential function with respect to θ from 0 to 2π .

Challenge 4: Analytical Derivation

1. **Transforming $\vec{F}$ into Polar Coordinates:** Substitute $x = r \cos \theta$ and $y = r \sin \theta$ into the force vector:

$$\vec{F} = -k(r \sin \theta \hat{i} - r \cos \theta \hat{j}) = kr(-\sin \theta \hat{i} + \cos \theta \hat{j})$$

Recognizing the polar unit vector $\hat{\theta} = -\sin \theta \hat{i} + \cos \theta \hat{j}$:

$$\vec{F} = kr\hat{\theta}$$

2. **Calculating Work Done:** The infinitesimal displacement vector is:

$$d\vec{s} = dr\hat{r} + r d\theta \hat{\theta}$$

Taking the dot product:

$$dW = \vec{F} \cdot d\vec{s} = (kr\hat{\theta}) \cdot (dr\hat{r} + r d\theta \hat{\theta})$$

Since $\hat{\theta} \cdot \hat{r} = 0$ and $\hat{\theta} \cdot \hat{\theta} = 1$:

$$dW = kr^2 d\theta$$

Now, substitute the equation of the logarithmic spiral $r(\theta) = R_0 e^{-a\theta}$:

$$dW = k (R_0 e^{-a\theta})^2 d\theta = k R_0^2 e^{-2a\theta} d\theta$$

We integrate this from $\theta = 0$ to $\theta = 2\pi$:

$$W = \int_0^{2\pi} k R_0^2 e^{-2a\theta} d\theta = k R_0^2 \left[\frac{e^{-2a\theta}}{-2a} \right]_0^{2\pi} = -\frac{k R_0^2}{2a} (e^{-4\pi a} - 1) = \frac{k R_0^2}{2a} (1 - e^{-4\pi a})$$

This is a positive, non-zero work, representing the energy transferred to the particle by the vortex-like force field along the spiral path.

Challenge 5: Frame-Dependence and Pseudo-Potential in a Rotating Frame

The Setup: A horizontal smooth guide track in the shape of a straight line rotates with a constant angular velocity ω about a vertical axis through the origin O . A bead of mass m is threaded on the track and is released from a distance r_0 from the rotation axis.

1. In the rotating frame of reference of the track, a centrifugal pseudo-force acts on the bead, given by $\vec{F}_{cf} = m\omega^2 r \hat{r}$. Show that this centrifugal force is conservative in this frame, and find its potential energy function $U_{cf}(r)$ taking $r = 0$ as the reference.
2. Show that the work done by the normal contact force exerted by the track on the bead is strictly zero in the rotating frame, but non-zero in the ground frame as the bead slides outwards.

Challenge 5: Progressive Guidance

- **Hint 1 (Centrifugal Potential):** The centrifugal force is purely 1D along the radial track. Use $dU_{cf} = -F_{cf} dr = -m\omega^2 r dr$ to find the potential function.
- **Hint 2 (Normal Force in Rotating Frame):** In the rotating frame, what is the direction of the displacement of the bead? What is the direction of the normal force? Look at their dot product.
- **Hint 3 (Coriolis and Normal Force in Ground Frame):** In the ground frame, the track is moving horizontally with speed $v_{\text{track}} = \omega r$. Therefore, the displacement of the bead has a horizontal azimuthal component as it slides radially outwards. The normal force must push the bead azimuthally to increase its angular momentum. Show that their dot product is positive.

Challenge 5: Analytical Derivation

1. **Centrifugal Potential:** The centrifugal force acts radially outward: $\vec{F}_{cf} = m\omega^2 r \hat{r}$. Since it is a 1D function of coordinate r , it is conservative by default.

$$dU_{cf} = -\vec{F}_{cf} \cdot d\vec{r} = -m\omega^2 r dr$$

Integrating from $r = 0$ where $U_{cf}(0) = 0$:

$$U_{cf}(r) = - \int_0^r m\omega^2 r' dr' = -\frac{1}{2} m\omega^2 r^2$$

This negative potential energy represents a "hill" in the rotating frame; as the bead moves outward (r increases), its potential energy decreases and its relative kinetic energy increases.

2. **Normal Force Work Comparison:**

- **In the Rotating Frame:** The bead is constrained to slide along the physical track, meaning its displacement vector $d\vec{s}'$ is strictly radial ($d\vec{s}' = dr \hat{r}$). The normal force $\vec{N}$ from the smooth track acts perpendicular to the track at all times ($\vec{N} = N \hat{\theta}$). Taking the dot product:

$$dW'_N = \vec{N} \cdot d\vec{s}' = (N \hat{\theta}) \cdot (dr \hat{r}) = 0 \implies \mathbf{W}'_N = 0$$

- **In the Ground Frame:** As the bead slides radially outward, the track is rotating. The actual displacement vector $d\vec{s}$ of the bead in the ground frame contains both radial and transverse components:

$$d\vec{s} = dr \hat{r} + r d\theta \hat{\theta}$$

Because the bead's distance r is increasing, its tangential speed $v_t = \omega r$ must also increase. This requires a transverse normal force to provide tangential acceleration (Coriolis effect in the ground frame):

$$N = 2m\omega \dot{r} \quad (\text{pointing in the direction of rotation, } \hat{\theta})$$

Taking the dot product to find the ground-frame work:

$$dW_N = \vec{N} \cdot d\vec{s} = (N \hat{\theta}) \cdot (dr \hat{r} + r d\theta \hat{\theta}) = Nr d\theta > 0 \implies \mathbf{W}_N > 0$$

Thus, the normal contact force does ****real, positive work**** in the ground frame to increase the bead's tangential kinetic energy, whereas it does ****strictly zero work**** in the co-rotating frame!